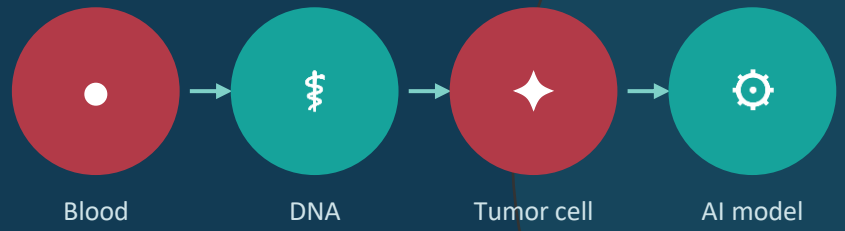
BSc CSE | MID / PROGRESS PRESENTATION

Explainable and Robust Machine Learning for Cancer Detection Using Plasma-Derived ctDNA Features

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Southeast University

Supervisor: [Supervisor Name]



Research Background

Existing Knowledge

- Cancer is one of the major global health challenges.
- Detecting cancer at an earlier stage is important in cancer management.
- Traditional diagnosis may involve imaging, tissue biopsy and laboratory tests.
- Tissue biopsy is invasive and samples one specific tumour location.
- Liquid biopsy offers a minimally invasive way to obtain molecular information.

Tissue biopsy



- Invasive procedure
- Single tumour site
- Hard to repeat often
- Needs tissue access

Liquid biopsy



- Blood-based sample
- Minimally invasive
- Can be repeated
- Molecular information

This research works only with already-published, de-identified public research data. No new patient sample is collected.

What Is Liquid Biopsy?

Existing Knowledge

Analysis of biological material circulating in body fluids — most commonly blood



i

Non-invasive / minimally invasive molecular information obtained from a routine blood draw.

- Plasma is separated from the blood sample in the laboratory.
- DNA fragments floating in plasma are extracted and sequenced.
- The sequencing output becomes the input for computational analysis.

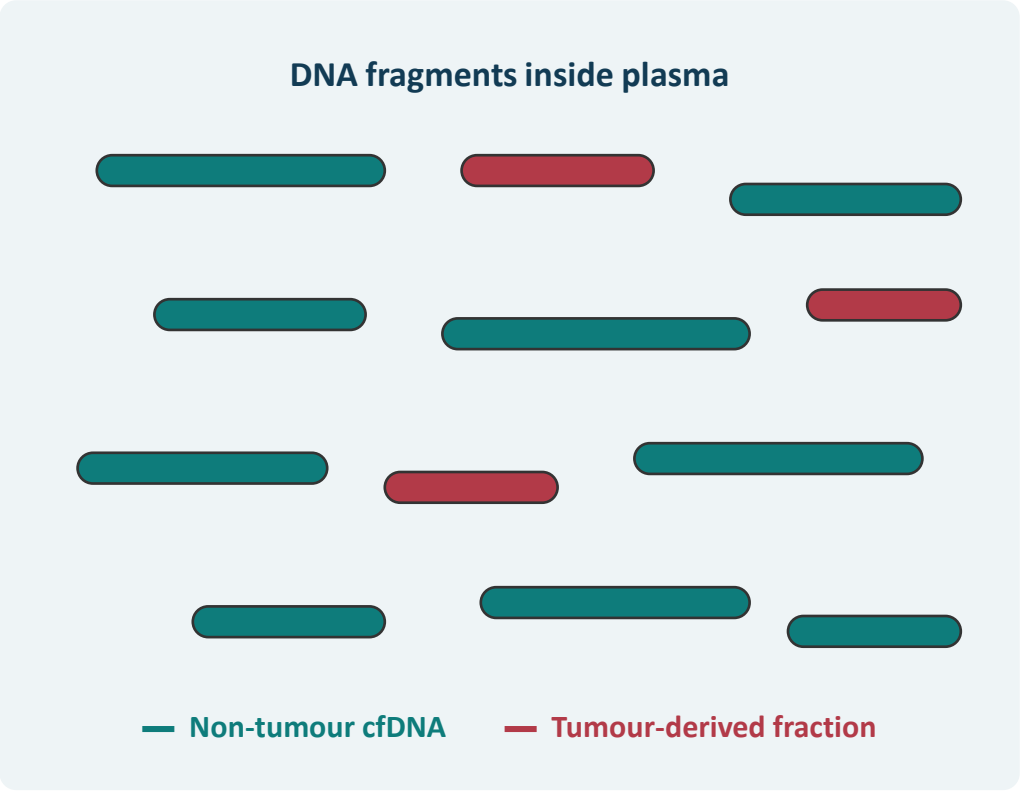
Scope note: this slide describes the general laboratory workflow reported in the liquid-biopsy literature; it does not claim any clinical performance.

What Is cfDNA?

Existing Knowledge

Cell-free DNA — DNA fragments circulating freely in blood plasma

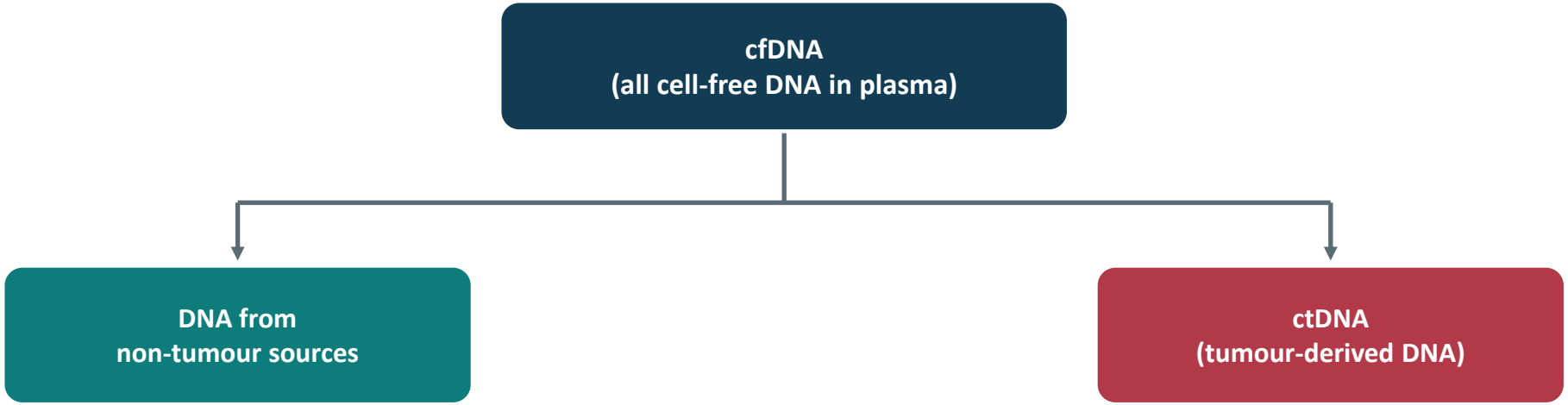
- cfDNA means cell-free DNA: short DNA fragments that are not inside a cell.
- These fragments can circulate in blood plasma.
- cfDNA can originate from many different cell types and biological sources.
- In cancer patients, a fraction of cfDNA can originate from tumour cells.
- cfDNA is usually present in small quantity, so measurement is technically sensitive.



Circulating Tumor DNA (ctDNA)

Existing Knowledge

ctDNA is a subset of cfDNA



- ctDNA = the tumour-derived fraction of cfDNA.
- Tumour-associated alterations can sometimes be detected in plasma-derived DNA.
- The ctDNA fraction is often very small compared with total cfDNA.
- This subset relationship is the single most important definition in my thesis.

- Carries tumour-associated genetic alterations
- Can reflect tumour molecular characteristics
- Main research target of liquid-biopsy studies

Why ctDNA?

Existing Knowledge

Potential advantages must be read together with real technical challenges

+ Potential advantages

- Blood-based sampling
- Minimally invasive for the patient
- Carries molecular information
- Can potentially be sampled repeatedly
- Numeric output, suitable for computational analysis

! Challenges

- Low abundance of ctDNA in plasma
- Large background of non-tumour cfDNA
- Biological variation between patients
- Technical noise and assay/sequencing effects
- Differences across cancer type and stage

These challenges are also the reason why a model can look accurate on one dataset and still be unstable — which is what my research examines.

What Information Can Be Obtained from ctDNA / cfDNA?

Existing Knowledge

1

Mutation features

Presence of tumour-associated variants

2

Mutant allele frequency

Proportion of mutant reads at a locus

3

Mutant fragments / mL

Mutant molecules per mL of plasma

4

DNA concentration

Amount of cfDNA in plasma

5

Copy-number alterations

Gains and losses across the genome

6

Fragmentomics

Genome-wide fragmentation profiles

7

Methylation features

Epigenetic marks on cfDNA

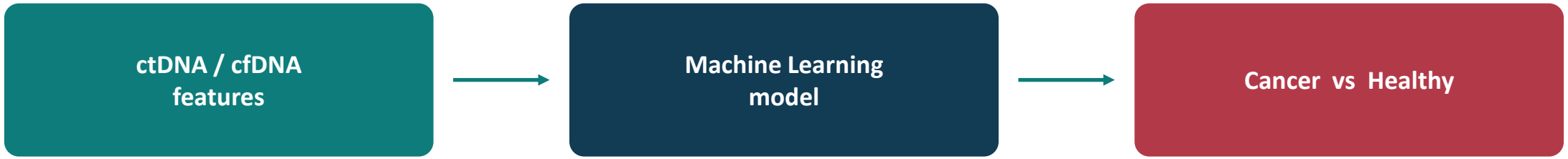
My initial focus: plasma-derived mutation / ctDNA-related features

Not every feature category is available in every dataset. The features I can actually use depend on what the selected public dataset provides.

Research Problem

Proposed Method

“Can machine learning identify patterns in plasma-derived ctDNA / cfDNA features that distinguish cancer samples from healthy controls?”



- This is a supervised binary classification task on a research dataset.
- Labels come from the published dataset, not from my own medical judgement.
- This is computational cancer classification, NOT a clinical diagnostic system.

0 = Healthy 1 = Cancer

Existing Research: CancerSEEK (Cohen et al., 2018)

Existing Knowledge

1,005 patients with clinically detected, nonmetastatic cancer

812 healthy controls

8 cancer types

A multi-analyte blood test combining plasma mutation information with circulating protein biomarkers.

- Breast
- Colorectal
- Esophageal
- Liver
- Lung
- Ovarian
- Pancreatic
- Stomach

Cohen JD, et al. Detection and localization of surgically resectable cancers with a multi-analyte blood test. Science, 2018. DOI: 10.1126/science.aar3247 (PMID 29348365)

CancerSEEK Dataset Structure

Proposed Method

Published supplementary data accompanying Cohen et al. (2018)

S4	Clinical / histopathological characteristics
S5	**Plasma mutations**
S6	Protein concentrations
S7	Plasma vs tumour mutation concordance

S5 (plasma mutations) is the table most relevant to my ctDNA / plasma-mutation-based analysis.

Potential input features

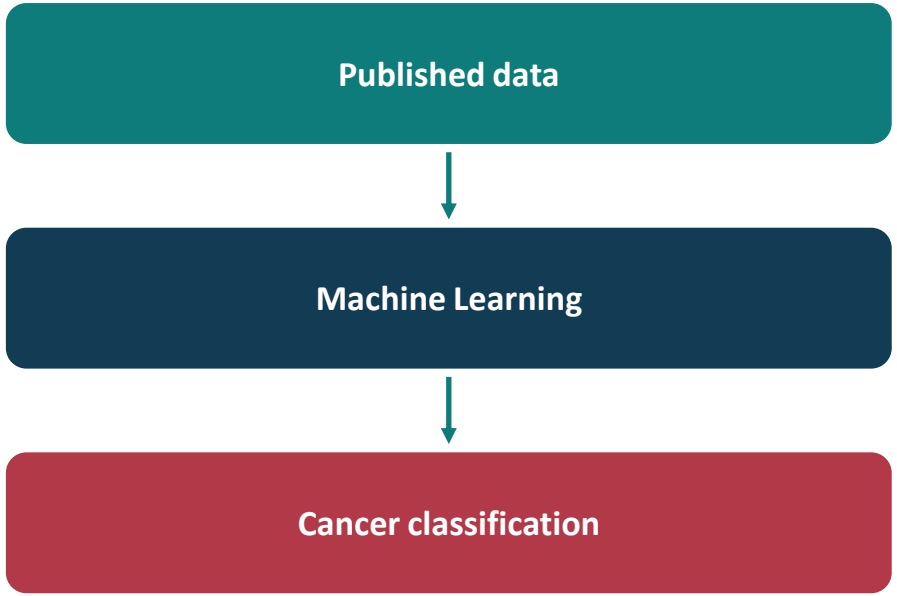
- Mutation information
- Omega score
- Mutant allele frequency (%)
- Mutant fragments per mL of plasma
- Plasma DNA concentration and plasma volume

✕ **The CancerSEEK Logistic Regression Score will NOT be used as an input feature — it is an output of the original CancerSEEK model, so using it would leak the answer into my model.**

Existing Machine Learning Research on This Data

Existing Knowledge

- Machine learning has already been applied to liquid-biopsy, cfDNA and ctDNA data.
- Wong et al. (2019) re-analysed the CancerSEEK data with several ML algorithms.
- CancerEMC (Rahaman et al., 2021) proposed a bagging ensemble meta-classifier with feature selection and oversampling on the same multi-analyte blood test data.
- DEcancer (Halner et al., 2023) added feature selection and a strict hold-out test set.
- Recent meta-analyses pool many ML + cfDNA studies across cancer types.

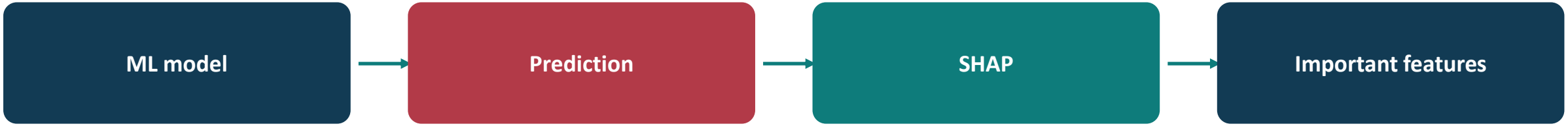


My research does NOT claim that applying machine learning to ctDNA data is new. That direction is already well established in the literature.

Wong KC et al., iScience 2019, DOI 10.1016/j.isci.2019.04.035 • Rahaman et al., CancerEMC, Bioinformatics 2021, DOI 10.1093/bioinformatics/btab044 • Halner A et al., DEcancer, iScience 2023, DOI 10.1016/j.isci.2023.106610

Existing Explainable AI (XAI) Research

Existing Knowledge



XAI answers one question

“Why did the model make this prediction?”

- SHAP — Shapley-value based, global and local
- LIME — local surrogate explanations
- Permutation importance — performance-drop based

- Interpretable ML and SHAP have already been used in recent liquid-biopsy and circulating-biomarker research.
- Example: SHAP-based interpretable multi-cancer detection on tumour-educated platelet RNA (2025).
- So I do NOT claim that nobody has used XAI with liquid-biopsy data.

Lundberg SM & Lee SI, NeurIPS 2017 (SHAP) • Ribeiro MT et al., KDD 2016 (LIME) • Interpretable multi-cancer early detection using SHAP-based ML on tumour-educated platelet RNA, PMC12427754

Identified Research Gap

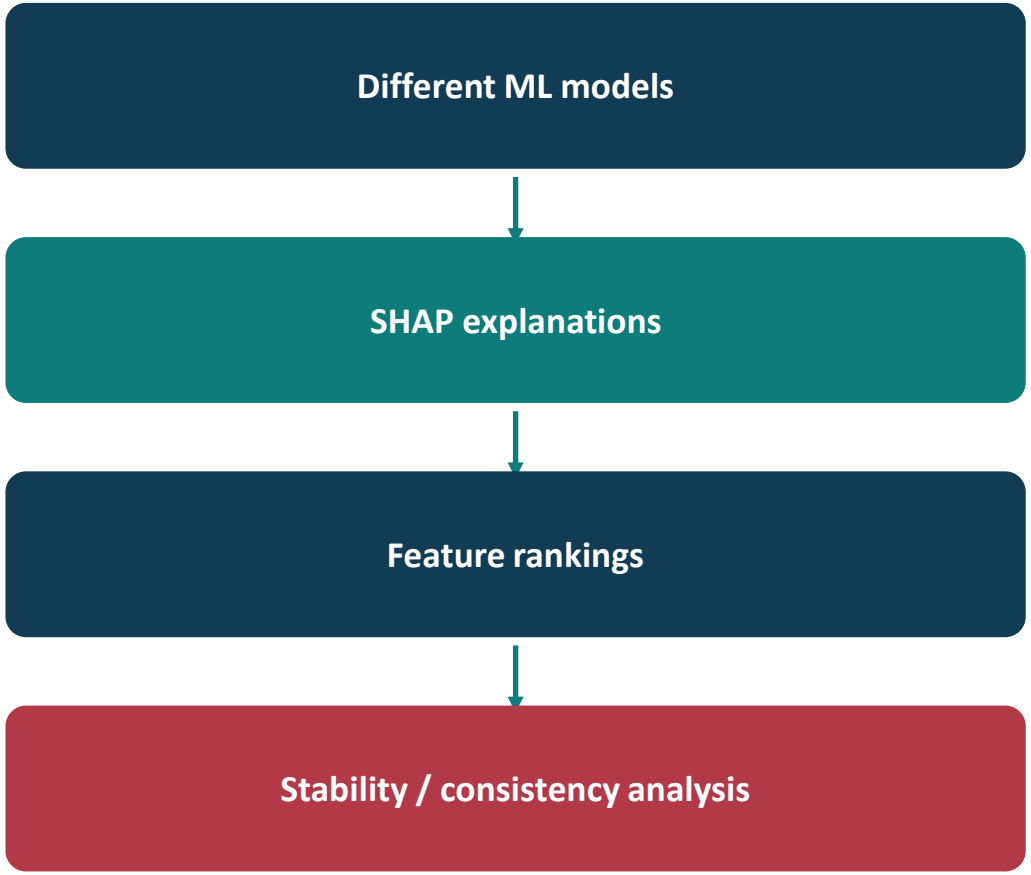
Research Gap

Potential methodological gap to investigate

- ML can be applied to ctDNA / cfDNA data — already shown.
- XAI can interpret those ML predictions — already shown.
- But different models, different folds and different datasets can produce different feature-importance rankings.

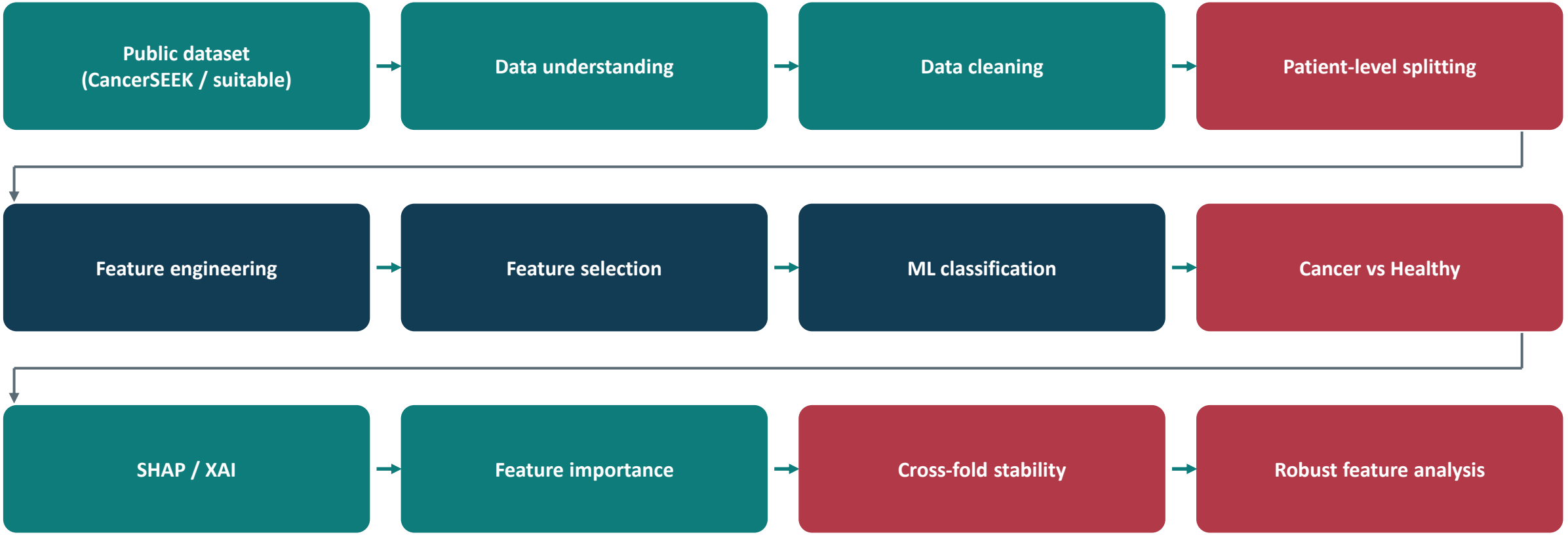
"Are XAI-identified important ctDNA-derived features stable and consistent across different ML models and validation splits?"

Wording used deliberately: this is a potential methodological gap to investigate, not a claim that no previous study has ever done this.



Proposed Research Framework

Proposed Method



Colour key: green = data preparation dark = modelling red = leakage control and robustness analysis

Input and Output

Proposed Method

INPUT

- Mutation features
- Omega score
- Mutant allele frequency (%)
- Mutant fragments / mL plasma
- Plasma DNA concentration
- Optional clinical features

MODELS

- Logistic Regression
- Random Forest
- XGBoost
- Support Vector Machine
- LightGBM

OUTPUT

Cancer vs Healthy

0 = Healthy

1 = Cancer

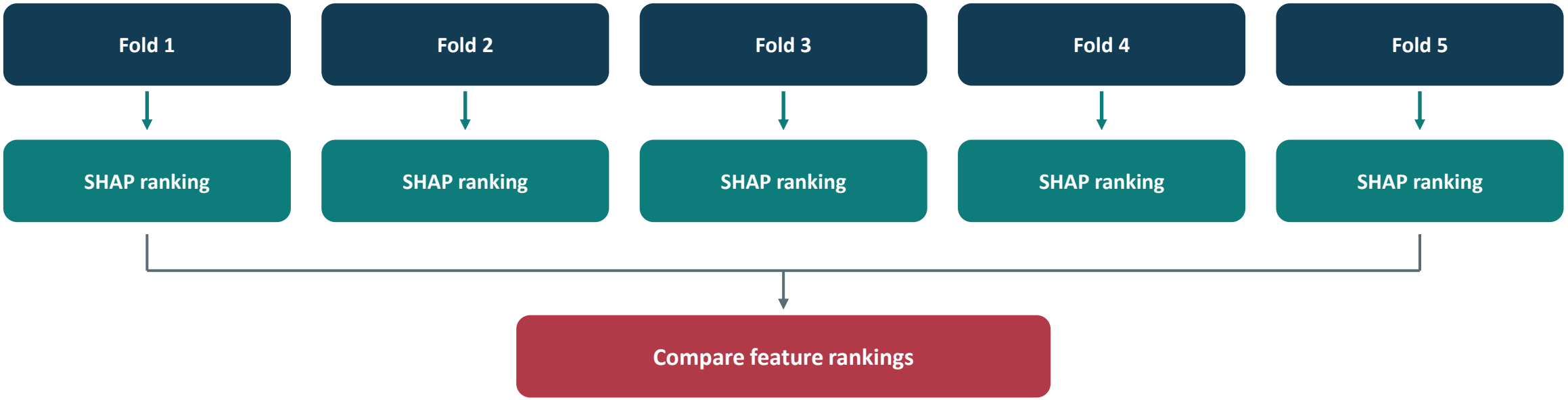
Binary classification, evaluated on a patient-level held-out test set.

The CancerSEEK Logistic Regression Score must NOT be used as an independent input feature — it is an output of the original published model.

XAI + Feature Stability Analysis

Proposed Method

The main proposed contribution of this thesis



Possible stability measures

- Feature selection frequency across folds
- Jaccard similarity of top-k feature sets
- Spearman and Kendall rank correlation

ILLUSTRATIVE EXAMPLE — NOT AN EXPERIMENTAL RESULT

Omega score → selected in 5 / 5 folds
Feature A → selected in 4 / 5 folds
Feature B → selected in 2 / 5 folds

Model Evaluation

Proposed Method

Accuracy
Overall correctness

Precision
Of predicted cancers, how many are right

Recall / Sensitivity
Of real cancers, how many are found

Specificity
Of healthy samples, how many are right

F1-score
Balance of precision and recall

ROC-AUC
Ranking quality across thresholds

PR-AUC
Better under class imbalance

MCC
Balanced single-number summary

Calibration / Brier
Are predicted probabilities realistic

Accuracy alone is not enough — especially when the classes are imbalanced.

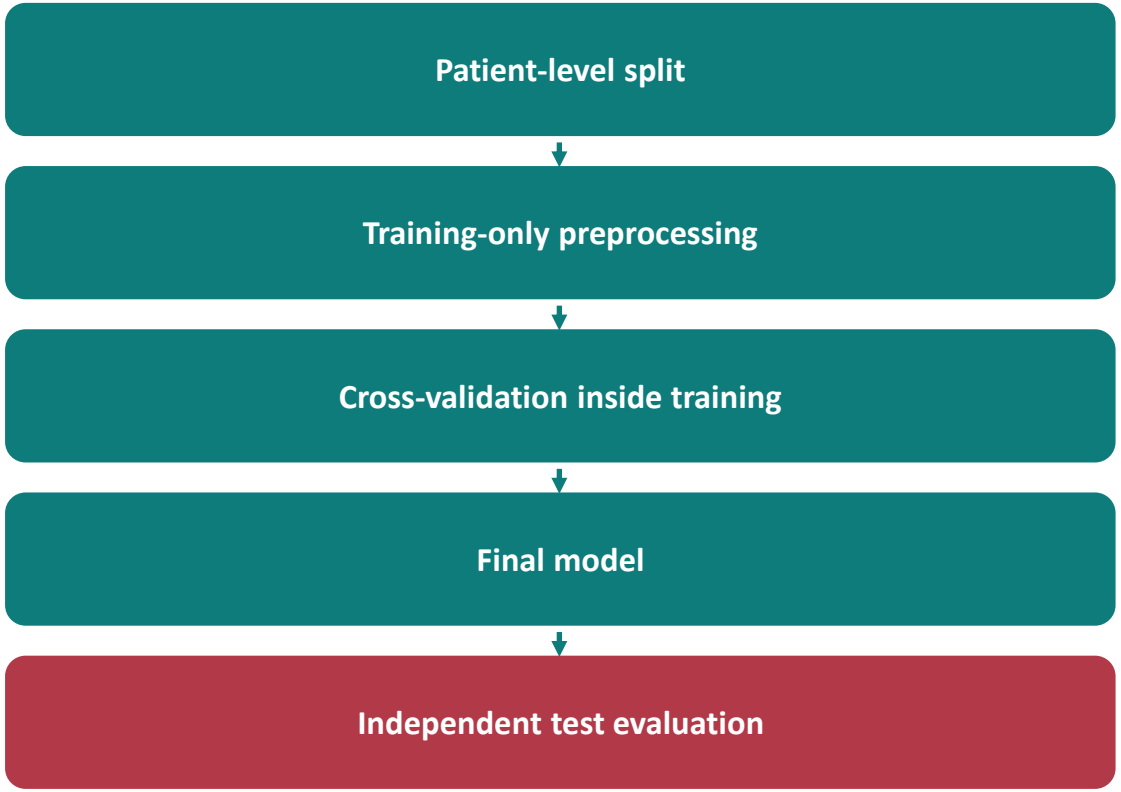
- Patient-level train / test split
- Cross-validation on the training portion only
- Final evaluation on an independent held-out test set

Data Leakage Control

Proposed Method

Potential leakage to avoid

- ✕ Same patient appearing in both train and test
- ✕ Feature selection performed before the split
- ✕ Oversampling performed before the split
- ✕ Using the original CancerSEEK model output as an input
- ✕ Using test data during preprocessing or scaling



NO PATIENT SHOULD APPEAR IN BOTH THE TRAINING AND THE TEST SET.

Expected Contribution

Expected Outcome

Expected Outcome — not an achieved result

1. A reproducible ctDNA / cfDNA-based machine learning pipeline
2. Comparative evaluation of multiple ML models on the same features
3. Explainable cancer classification using SHAP and other XAI methods
4. Identification of important plasma-derived molecular features
5. Evaluation of feature-importance stability across folds
6. Cross-model and cross-fold consistency analysis
7. Clear documentation of limitations and data-leakage controls

No experimental numbers are reported in this presentation. Accuracy, AUC, sensitivity and specificity will only be reported after the experiments are completed.

Limitations and Future Work

Future Work

Limitations

- Public dataset limitations
- Sample size and representativeness
- Dataset-specific bias
- Batch and assay effects
- Low ctDNA abundance
- Limited external validation
- Only the features the dataset provides are available

Future work

- External validation on an independent cohort
- Larger cohorts and more cancer types
- Independent public datasets
- Fragmentomics features
- DNA methylation features
- Copy-number alterations
- Multi-omics integration and early-stage focus

Conclusion

ctDNA	Provides tumour-derived molecular information from a blood sample
Machine Learning	Can learn patterns from those molecular features
Explainable AI	Can help interpret why the model made a prediction
Proposed research	Investigates prediction performance AND explanation stability

"From Molecular Signals to Explainable and Robust Predictions"

References

[1] Cohen JD, et al. Detection and localization of surgically resectable cancers with a multi-analyte blood test. Science, 2018;359(6378):926-930. DOI: 10.1126/science.aar3247 (PMID 29348365)

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[10] Value of machine learning models for cell-free DNA-based multi-cancer early detection: a systematic review and meta-analysis. PubMed PMID 41719115 (PMC12925023).

[11] Artificial intelligence-enabled liquid biopsy in cancer: a systematic review and meta-analysis of diagnostic performance and biological implications. PubMed PMID 42388633.

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